TITLE PAGE

Project Title: Library Management System (Java GUI)

Problem Statement:

Manual library management is time-consuming and error-prone. This system digitizes book handling operations.

Target Users:

- Librarians
- Students
- Educational Institutions

SYSTEM DESIGN

Architecture:

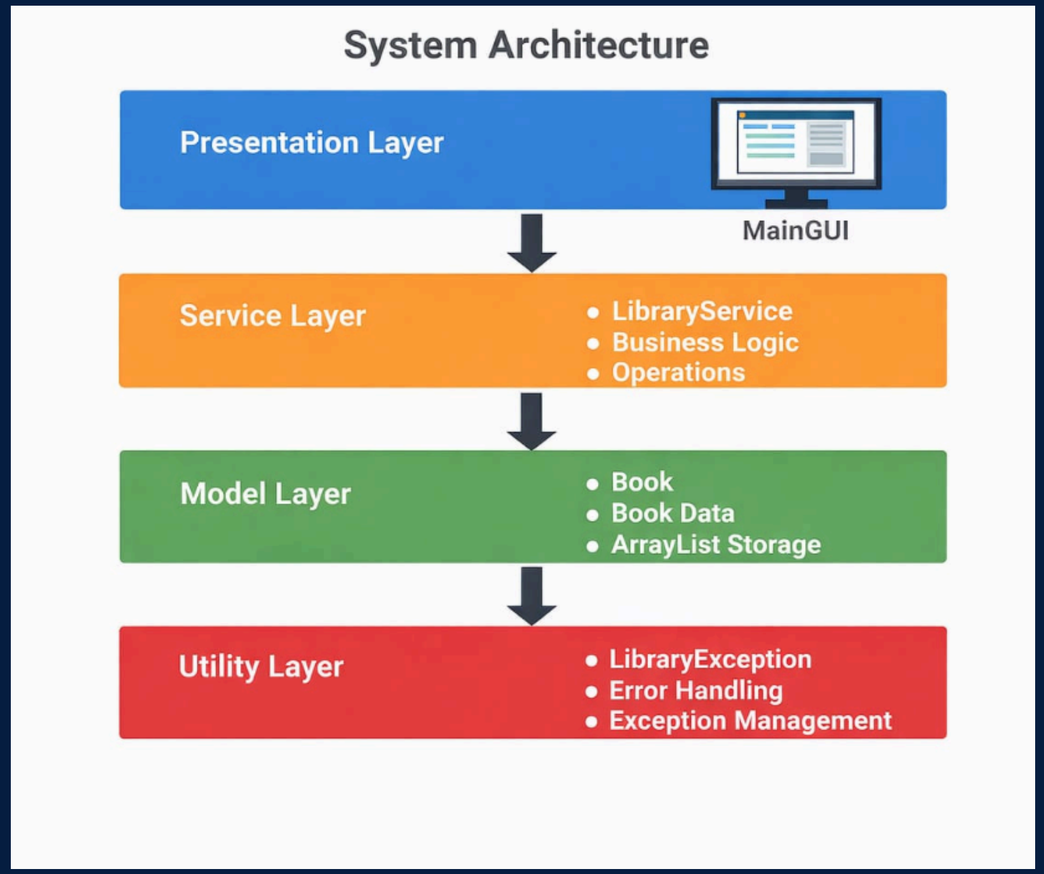
- Model → Book
- Service → LibraryService
- Util → LibraryException
- Main → MainGUI

Key Classes:

- Book → stores book data
- LibraryService → handles logic
- MainGUI → user interface
- LibraryException → error handling

Package Structure:

- model
- service
- util and main



OOP CONCEPTS

- **Encapsulation :**
  -Book class → private variables
- **Abstraction :**
  -LibraryService hides internal logic
- **Inheritance :**
  -LibraryException extends Exception
- **Polymorphism :**
  -Exception handling using try-catch

TECHNICAL HIGHLIGHT

- **Feature:**

Book Issue Logic

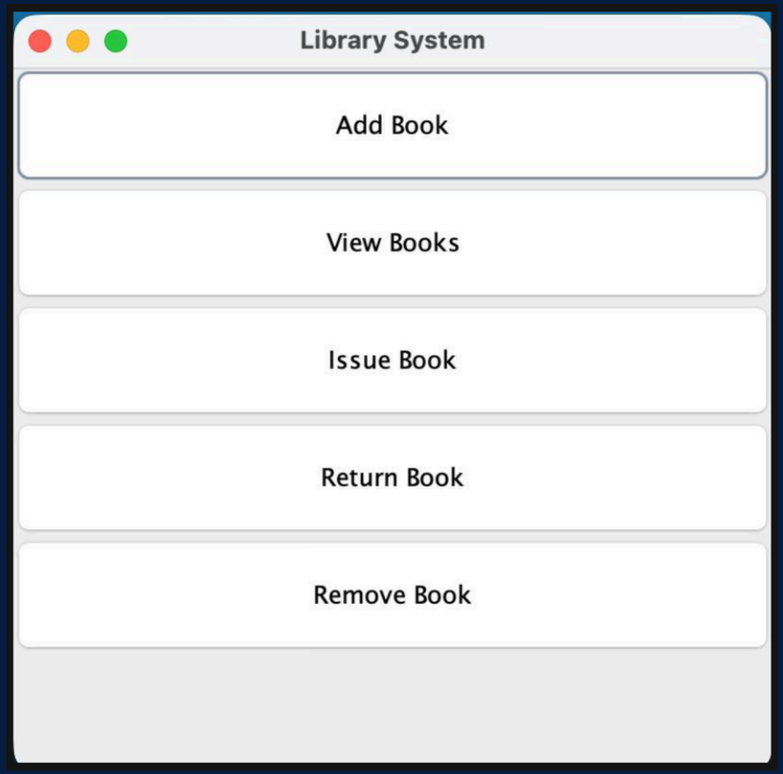
```
if (b.isIssued()) {
throw new LibraryException("Book already issued!");
}
```

- **Why?**

-Prevents duplicate issuing
-Ensures data consistency
-Uses exception handling

LIVE DEMO

- Step 1: Add Book
- Enter ID, title, author
- Step 2 : View Books, Show list
- Step 3: Issue Book
- Show status change
- Step 4: Error Case
- Issue same book again → error



THANK YOU

AUGNIK KUMAR SAHOO 12024052017056 CSE(IOT-CS-BT) ROLL-54